What Predicted the 2008 Housing Crash?

Bubble Deflation vs. Economic Fundamentals

Introduction

The 2008 financial crisis produced the largest decline in U.S. home prices since the Great Depression, but the damage was not evenly distributed. Nevada lost over 55% of its peak home value, while North Dakota lost nothing. This raises a central question: was the crash simply a bubble deflation, where states that appreciated the most fell the hardest, or did pre-crisis economic fundamentals determine which states were resilient?

We use state-level data from the Federal Reserve Economic Data (FRED) platform: four time series per state (house price index, unemployment rate, per-capita income, population) assembled into a cross-sectional dataset of 51 observations. The outcome variable is the peak-to-trough percentage decline in the FHFA All-Transactions House Price Index. All predictors are pre-crisis indicators observable before the crash. Prior work has focused on loan-level data (subprime mortgage share, loan-to-value ratios). Our approach uses publicly available macro-level indicators, tested through model comparison, a permutation simulation, and an experiment comparing OLS regression to an LLM performing the same prediction task with a contamination control.

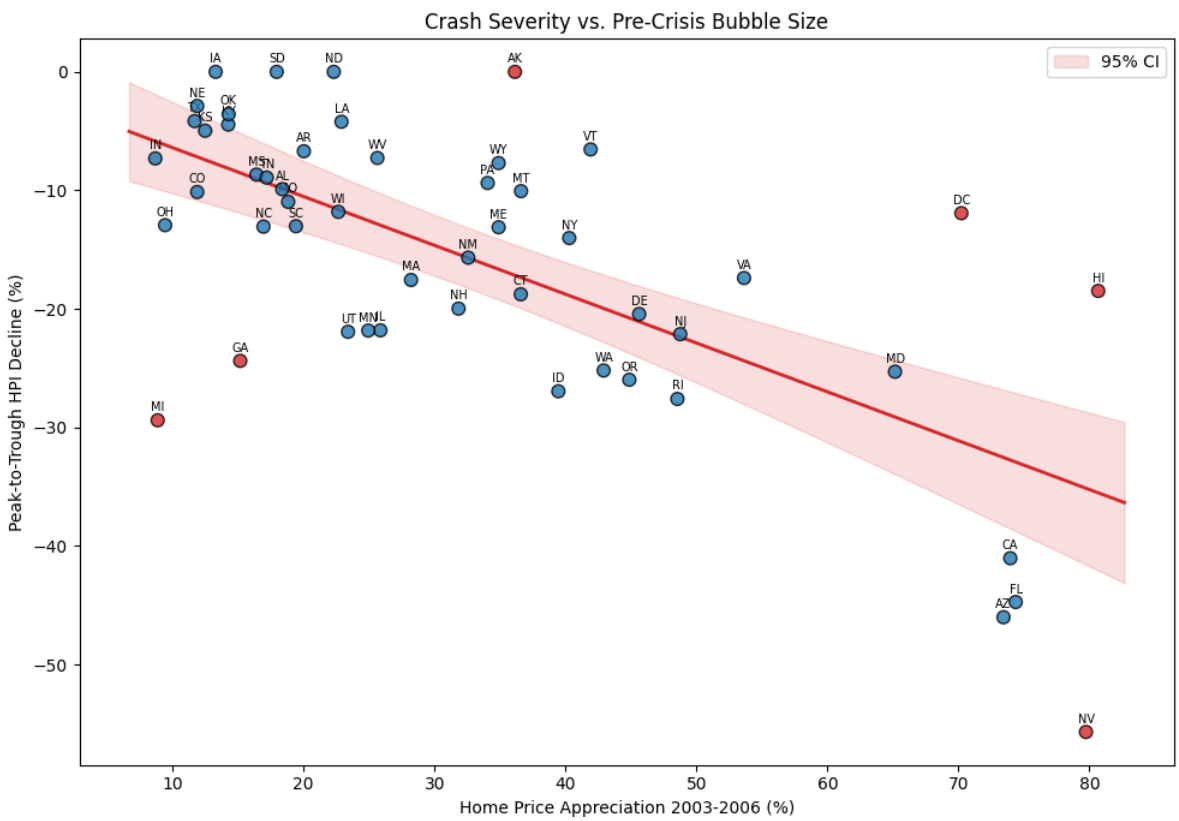
Data wrangling

The raw data consists of 204 CSV files downloaded from FRED, four time series for each of the 50 states plus DC spanning 2000-2012. We read each series, extract summary statistics at key dates, and assemble them into a single cross-sectional DataFrame with one row per state. Full data construction code is available in the project repository.

Computing crash severity from the House Price Index

For each state, we find the peak HPI value in the 2000-2012 window, then find the minimum HPI after that peak date. The crash percentage is `(trough - peak) / peak * 100`. We also compute pre-crisis appreciation as the percent change in HPI from Q1 2003 to Q1 2006, which measures how much the bubble inflated before it burst.

Visualizations



`r = -0.684, R-squared = 0.467`

Analysis: does the bubble explain the crash?

```
lm_simple = smf.ols('crash_pct ~ appreciation_03_06', data=df).fit()
print(f'R-squared: {lm_simple.rsquared:.3f}')
print(pd.DataFrame({'coef': lm_simple.params, 'std err': lm_simple.bse,
 ↪  'p-value': lm_simple.pvalues}).round(4))
```

```
R-squared: 0.467
                      coef  std err  p-value
Intercept          -2.2788   2.4212   0.3512
appreciation_03_06 -0.4120   0.0629   0.0000
```

```
lm_multi = smf.ols('crash_pct ~ appreciation_03_06 + ur_2006 + pcpi_2006 +
 ↪  pop_growth_03_06', data=df).fit()
print(f'R-squared: {lm_multi.rsquared:.3f}')
print(pd.DataFrame({'coef': lm_multi.params, 'std err': lm_multi.bse,
 ↪  'p-value': lm_multi.pvalues}).round(4))
```

```
R-squared: 0.582
                       coef  std err  p-value
Intercept           10.3390  10.5473   0.3321
appreciation_03_06  -0.3765   0.0751   0.0000
ur_2006             -2.2752   1.2133   0.0671
pcpi_2006            0.0000   0.0002   0.9490
pop_growth_03_06    -1.4420   0.4845   0.0046
```

```
from statsmodels.stats.anova import anova_lm
print(anova_lm(lm_simple, lm_multi))
```

```
   df_resid          ssr  df_diff     ss_diff         F    Pr(>F)
0      49.0  4012.335552      0.0         NaN       NaN       NaN
1      46.0  3151.414654      3.0  860.920898  4.188845  0.010558
```

Model 1 (appreciation only) explains about 47% of the variance in crash severity. Each additional percentage point of pre-crisis appreciation corresponds to roughly a 0.41 percentage point larger crash. Model 2 adds unemployment, income, and population growth, raising R-squared to about 0.58. The F-test above evaluates whether the three additional predictors are jointly significant, that is, whether Model 2 explains significantly more variance than Model 1. If the p-value is below 0.05, the economic fundamentals add real predictive power beyond the bubble alone.

Simulation: is the relationship real?

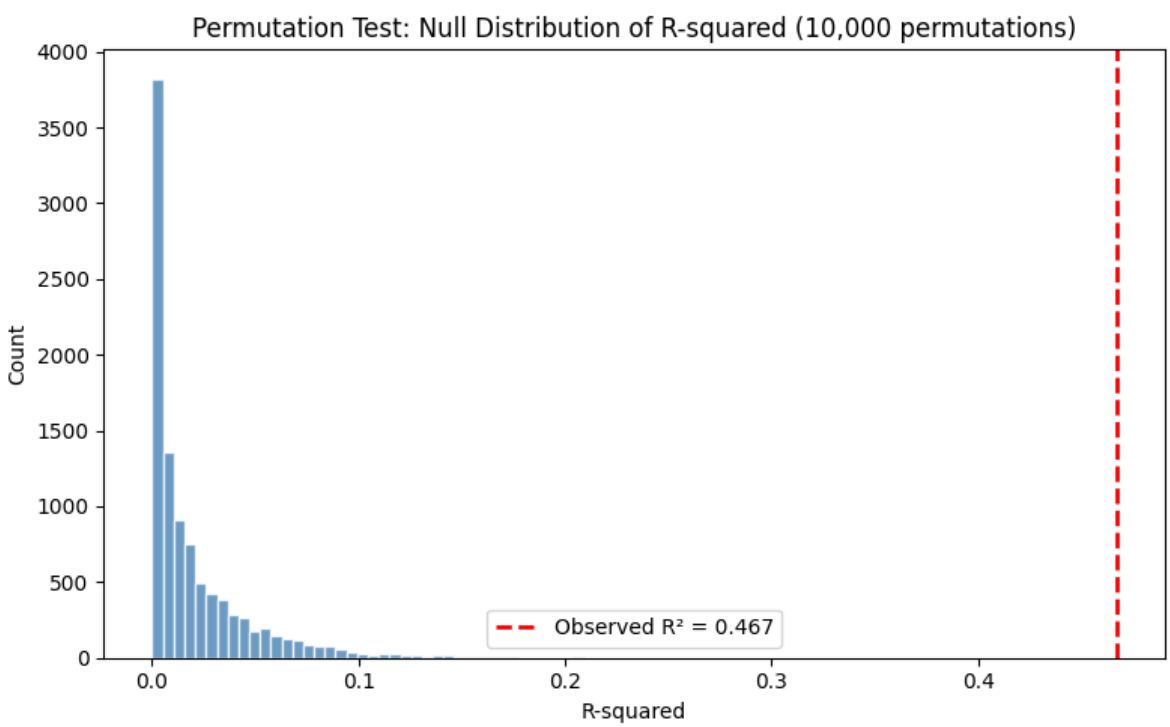
Under the null hypothesis that pre-crisis appreciation has no relationship with crash severity, we would expect any apparent correlation to vanish if we shuffled the crash labels. To test this, we permute `crash_pct` across states 10,000 times, refit the simple regression each time, and collect the null distribution of R-squared values. We then compare our observed R-squared to this distribution.

```python
np.random.seed(42)
n_perm = 10_000
null_r2 = np.empty(n_perm)

for i in range(n_perm):
    df_perm = df.copy()
    df_perm['crash_pct'] = np.random.permutation(df['crash_pct'].values)
    m = smf.ols('crash_pct ~ appreciation_03_06', data=df_perm).fit()
    null_r2[i] = m.rsquared

observed_r2 = lm_simple.rsquared
p_value = np.mean(null_r2 >= observed_r2)
print(f'Observed R-squared: {observed_r2:.4f}')
print(f'Permutation p-value: {p_value:.4f}')
```

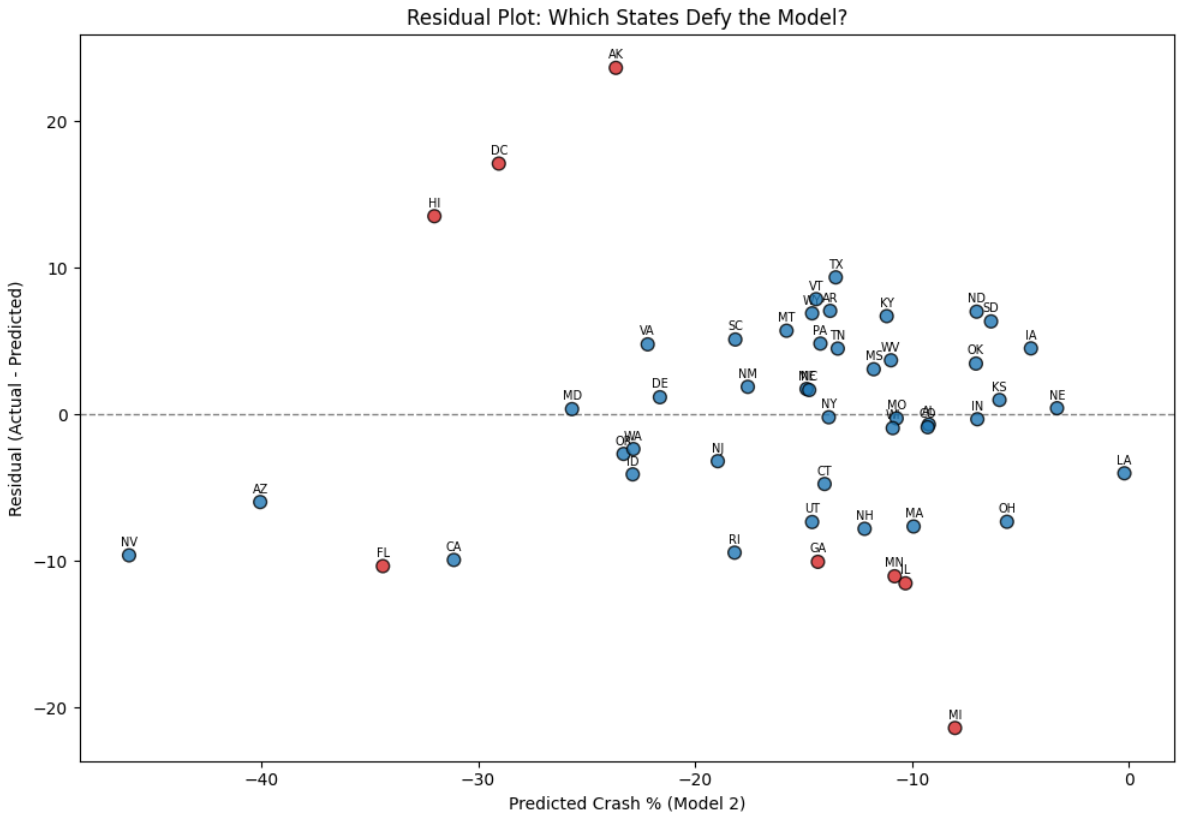
```
Observed R-squared: 0.4672
Permutation p-value: 0.0000
```



The observed R-squared is far above the null distribution, confirming that the relationship between pre-crisis appreciation and crash severity is not an artifact of chance. The bubble-crash link is real.

Which states defy the model?

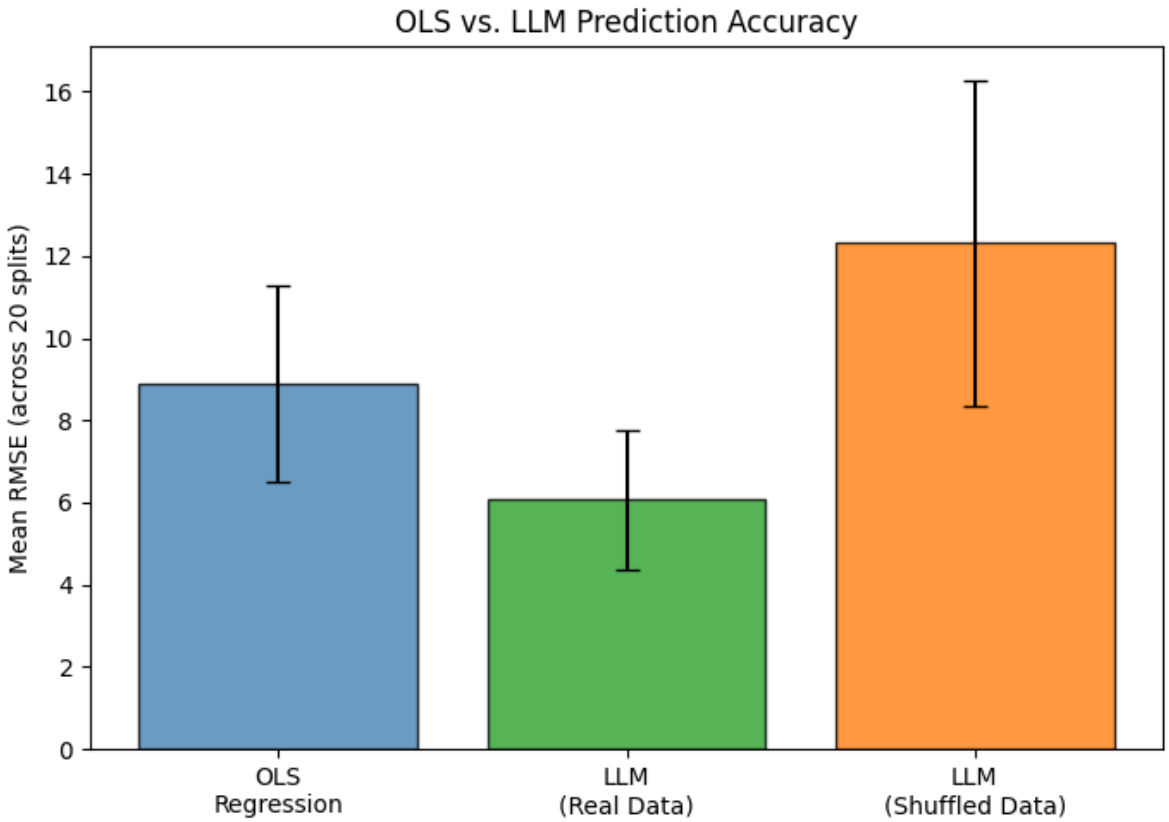
The regression captures the general pattern, but some states crash much harder or softer than the multiple regression predicts. These outliers reveal what the model cannot capture: idiosyncratic economic structure that four pre-crisis indicators do not reflect.



Can an LLM match a regression?

Large language models encode economic knowledge from their training data. We test whether Gemini Flash, given the same training data as OLS, can predict held-out states' crash severity. This directly compares statistical inference (OLS) to in-context learning (LLM) on the same small-data task.

The LLM may have memorized 2008 crisis outcomes during pre-training. To detect this, we run a parallel condition where the training outcomes are randomly shuffled. If the LLM performs equally well on real and shuffled training data, it is relying on prior knowledge rather than learning from the provided data. If it performs worse on shuffled data, in-context learning is active.




```
Mean RMSE - OLS:            8.88 (+/- 2.39)
Mean RMSE - LLM (real):     6.06 (+/- 1.68)
Mean RMSE - LLM (shuffled): 12.31 (+/- 3.97)
```

The LLM with real training data (mean RMSE 6.06) outperforms OLS regression (mean RMSE 8.88) in 17 of 20 splits. The contamination control confirms that in-context learning is active: when training outcomes are shuffled, the LLM's RMSE rises to 12.31, worse than both OLS and the real-data condition, in 19 of 20 splits. This rules out the possibility that the LLM is simply recalling memorized crisis outcomes. Instead, the LLM extracts genuine signal from the provided training data and combines it with prior economic knowledge encoded during pre-training. When both sources agree (real data condition), the LLM dominates OLS. When they conflict (shuffled condition), performance degrades sharply.

Conclusions

The 2008 housing crash was primarily a bubble deflation, but not entirely. Pre-crisis home price appreciation alone explains 47% of the cross-state variance in crash severity (R-squared = 0.47, $p < 0.0001$). Each additional percentage point of appreciation from 2003 to 2006

corresponds to a 0.41 percentage point larger crash. The permutation test confirms this is not an artifact of chance: in 10,000 random shuffles, no permuted R-squared reached the observed value.

Adding economic fundamentals raises R-squared to 0.58, and the F-test confirms the improvement is significant ($p = 0.011$). Population growth is the strongest additional predictor ($p = 0.005$): states with rapid pre-crisis in-migration faced larger crashes. Unemployment is marginally significant ($p = 0.067$), while income has no detectable effect. The bubble is the dominant story, but fundamentals, particularly population dynamics, meaningfully shape which states were resilient.

The residual analysis reveals what four pre-crisis indicators cannot capture. Michigan crashed 21 percentage points harder than predicted, reflecting its auto industry collapse. Alaska, DC, and Hawaii crashed far less than expected, insulated by government employment, military presence, and geographic isolation.

The LLM experiment produced a striking result. Gemini Flash achieved a mean RMSE of 6.06 compared to 8.88 for OLS, winning 17 of 20 splits. The shuffled control (mean RMSE 12.31) confirms genuine in-context learning: the LLM performs much worse when training labels are scrambled. The LLM combines statistical signal from the provided data with prior economic knowledge, outperforming simple regression on this small-data task. This suggests that for cross-sectional problems with few observations but rich domain context, in-context learning may complement traditional statistical methods.

Reflection

To be completed.

Appendix

Rendered notebook with full code: final_project.ipynb

Source code and data: github.com/ethanaggor/sds1730-final-project